

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Placebo-controlled Phase III Study to Evaluate the Efficacy and Safety of Dabrafenib Plus Trametinib in Previously Treated Patients With Locally Advanced or Metastatic, Radio-active Iodine Refractory BRAFV600E Mutation-positive Differentiated Thyroid Cancer (DTC) **Short Title/Acronym:** Efficacy and Safety of Dabrafenib Plus Trametinib in BRAFV600E-mutant DTC **Protocol Number:** CDRB436J12301 **NCT Number:** NCT04940052 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Dabrafenib (Tafinlar) + Trametinib (Mekinist) **Phase of Development:** Phase III **Trial Status:** Active, not recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the Progression-Free Survival (PFS) of dabrafenib plus trametinib compared to placebo. **Secondary Objectives:** To evaluate Overall Response Rate (ORR), Overall Survival (OS), Duration of Response (DOR), and the safety and tolerability profile of the combination therapy.

2.2 Background & Rationale To address the clinical need for effective targeted therapies in patients with locally advanced or metastatic BRAF V600E mutation-positive, differentiated thyroid carcinoma (DTC) who are refractory to radioactive iodine (RAI) and have progressed following prior VEGFR targeted therapy. While multikinase inhibitors are standard frontline therapies, patients often experience disease progression. This study evaluates if combined BRAF and MEK inhibition (dabrafenib + trametinib) can significantly prolong progression-free survival and improve response rates in this specific patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (2:1 ratio to dabrafenib + trametinib vs. placebo)

3.2 Planned Enrollment & Duration Target \$\$\$: 153 patients **Number of Sites:** Global, multicenter (United States + 10 other countries) **Estimated Study Start:** 14-Nov-2021 **Estimated Primary Completion:** 03-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female ≥ 18 years of age at the time of informed consent. 2. Histologically or cytologically confirmed diagnosis of advanced/metastatic differentiated thyroid carcinoma. 3. Radioactive-iodine refractory disease. 4. BRAF V600E mutation-positive tumor sample as per central laboratory result. 5. Has progressed on at least 1 but not more than 2 prior VEGFR targeted therapies. 6. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2 . 7. At least one measurable lesion as defined by RECIST v1.1.

4.2 Exclusion Criteria 1. Anaplastic or medullary carcinoma of the thyroid. 2. Previous treatment with a BRAF inhibitor and/or a MEK inhibitor. 3. Concomitant RET Fusion-Positive Thyroid Cancer. 4. Treatment with any type of small molecule kinase inhibitor (including investigational kinase inhibitor) within 2 weeks before randomization. 5. Treatment with any type of anticancer antibody (including investigational antibody) or systemic chemotherapy within 4 weeks before randomization. 6. Treatment with radiation therapy for bone metastasis within 2 weeks or any other radiation therapy within 4 weeks before randomization. 7. A history or current evidence/risk of retinal vein occlusion (RVO) or central serous retinopathy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dabrafenib 150 mg twice a day (BID) and Trametinib 2 mg once a day (QD). The control arm receives matching placebos. **Route of Administration:** Oral **Duration of Treatment:** Until disease progression as per RECIST 1.1 (confirmed by BIRC), unacceptable toxicity, loss of clinical benefit, withdrawal of consent, or death. Note: Patients in the placebo arm with confirmed progression have the option to cross over to the open-label active treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications. **Prohibited:** Concurrent anti-cancer therapies, other BRAF/MEK inhibitors, unauthorized small molecule kinase inhibitors, and concurrent radiation therapy (unless specified wash-out periods are met prior to randomization).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Tumor BRAF Mutation Testing, Baseline Imaging
Baseline	Day 0	Randomization (2:1), First Dose of Study Drug or Placebo	Interim Every 4-8 Weeks Safety Labs, Efficacy Assessment (Tumor Imaging via RECIST 1.1) End of

Study | Varies | End of treatment due to progression or toxicity, Crossover evaluation (for Placebo arm), Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from the date of randomization to the date of the first documented progression according to RECIST 1.1 based on Blinded Independent Review Committee (BIRC) assessment, or death due to any cause.

7.2 Secondary & Exploratory Endpoints * Overall Response Rate (ORR) * Overall Survival (OS) * Duration of Response (DOR) * Incidence and severity of Adverse Events (AEs) to determine safety and tolerability

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard safety monitoring encompassing laboratory AEs (e.g., anemia, hypoglycemia, decreased neutrophil/white blood cell counts) and clinical AEs. **Serious Adverse Events (SAEs):** Pyrexia (fever) is a noted common treatment-related SAE requiring standard 24-hour reporting protocols. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) for overall safety, alongside the BIRC for independent efficacy/progression review.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints (PFS, ORR, OS, DOR). Safety Set for all patients receiving at least one dose of the study drug/placebo. **Sample Size Justification:** $N = 153$, randomized 2:1, powered to detect a statistically significant hazard ratio (HR) superiority of the dabrafenib + trametinib combination over placebo for PFS. **Handling of Missing Data:** Standard right-censoring methods for time-to-event endpoints (PFS/OS) and non-responder imputation for ORR where applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard risk-based onsite and remote monitoring of clinical sites to ensure protocol adherence and patient safety. **Audit Trail:** Validated eCRF locking processes, coupled with strictly controlled data transfer workflows for the central laboratory (BRAF testing) and BIRC imaging assessments.

11. Study Identifiers & Governance

Primary ID: CDRB436J12301 **Posting ID:** 812737524117083827 **Registry Number:** NCT04940052 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Study of Efficacy and Safety of Dabrafenib in Combination With Trametinib in Previously Treated Patients With Metastatic, Radio-active Iodine Refractory BRAF V600E Mutation Positive Differentiated Thyroid Cancer **Official Regulatory Title:** A Randomized, Double-blind, Placebo-controlled Phase III Study to Evaluate the Efficacy and Safety of Dabrafenib Plus Trametinib in Previously Treated Patients With Locally Advanced or Metastatic, Radio-active Iodine Refractory BRAFV600E Mutation-positive Differentiated Thyroid Cancer (DTC)

12. Clinical Framework (Oncology Specific)

Primary Condition: Differentiated Thyroid Cancer (DTC) **Diagnosis Threshold:** Radio-active Iodine (RAI) Refractory AND BRAF V600E Mutation-positive **Medical Methodology:** Targeted combination kinase inhibition **Optimization Target:** Progression-Free Survival (PFS) prolongation

13. Study Procedures & Methodology

Management Pathway: 2:1 Randomization (Active vs. Placebo) with an option for placebo patients to cross over to open-label Dabrafenib + Trametinib upon BIRC-confirmed progression. **Efficacy Assessment:** Centralized imaging evaluation via BIRC. **Safety Protocol:** Continuous tracking of known MEK/BRAF inhibitor toxicities (e.g., pyrexia, cytopenias).

14. Observation & Follow-Up Schedule

Total Duration: From screening until disease progression, unacceptable toxicity, or death. **Onsite Visit Cadence:** Routine oncology schedule (e.g., Q4W to Q8W) for drug dispensing and tumor assessments. **Radiographic Monitoring Frequency:** According to RECIST 1.1 interval requirements. **Survival Follow-up Interval:** Every 12 weeks following treatment discontinuation until death or study closure.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				